Udit Agarwal

United States | +1-930-904-4901 | agarwaludit13@gmail.com | linkedin.com/in/udit013 | github.com/Udit013 | uditagarwal.vercel.app

EDUCATION

Indiana University Bloomington — Bloomington, Indiana, US
Master of Science in Computer Science, GPA: 3.84/4.0 — Aug 2024 - May 2026

Kalinga Institute of Industrial Technology (KIIT University) — Bhubaneswar, Odisha, India
Bachelor of Technology in Computer Science and Engineering, GPA: 8.85/10.0 — Aug 2020 - May 2024

EXPERIENCE

Heartland Community Network — Jun 2026 - Present
Senior Consultant — Remote, Indiana, US

- Architect and deliver full-stack applications, backend services, and REST APIs against client-specific operational requirements, owning design, implementation, testing, deployment, and ongoing maintenance
- Engineer AI-powered features and LLM-driven automation workflows that cut manual process steps, speed up information retrieval, and improve data-driven decision-making across client industries
- Translate complex and ambiguous client requirements into scalable technical designs, then own delivery through code review, testing, deployment, and maintenance

Global Health Impact Project — Jan 2026 - Jun 2026
Software Engineer — Indiana University, Bloomington, Indiana

- Engineered full-stack analytics features in React, TypeScript, and Python enabling researchers to evaluate pharmaceutical intervention coverage, efficacy, and population-level outcomes
- Built backend services and data pipelines for treatment-coverage modeling, efficacy analysis, and large-scale population health forecasting, exposed through REST APIs to forecasting dashboards
- Identified and resolved application bottlenecks and production issues across the stack, optimizing database queries and backend processing to support increasingly complex analytical workloads

University Information Technology Services (UITS) — Aug 2025 - Present
Consultant - IT Services — Indiana University, Bloomington, Indiana

- Resolve 100+ technical issues weekly across desktop, mobile, and enterprise applications through structured troubleshooting and root-cause analysis, documenting resolutions in ServiceNow
- Support enterprise identity and access management across authentication, authorization, account provisioning, and lifecycle workflows for a large university user base

PROJECTS

PrepWise - AI Voice Interview Platform | *Next.js, TypeScript, Firebase, Web Speech API, Monaco Editor, Pyodide/WASM, Gemini, Zod*

- Shipped live coding rounds with real in-browser execution for JavaScript and Python, killing infinite loops at a 5-second (JS) and 15-second (Python) timeout without freezing the tab, by running code in a terminable Web Worker with a lazily loaded Pyodide WASM runtime
- Cut per-interview speech cost to zero across 5 interview formats with browser-native recognition and synthesis, rebuilding capture as continuous recognition that accumulates segments across pauses and recovers from recognizer shutdowns
- Kept the progress dashboard to 173 bytes of route-level JavaScript and halved its Firestore reads with server-rendered SVG charts, raised secondary-text contrast from 3.6:1 to 5.7:1 to meet WCAG AA, all covered by a 172-test Vitest suite in GitHub Actions CI
- Raised free AI capacity from 20 to 1,000 requests per day and cut interview turns from about 4-7 seconds to 2-2.5 seconds by moving to Gemini Flash-Lite behind a model chain that fails over to a second model on quota exhaustion or model retirement

CipherWatch - Credential Exposure Ledger | *Python, FastAPI, SQLAlchemy (async), PostgreSQL, WebSockets, APScheduler, Docker, pytest*

- Delivered an async FastAPI backend of 22 HTTP endpoints plus a WebSocket, with an optional private mode whose token rides in Sec-WebSocket-Protocol so it never reaches URLs or logs, and per-IP sliding-window rate limits of 10 scans per 10 minutes
- Found and fixed real defects by testing against PostgreSQL and real repositories: timezone-aware scheduler timestamps that Postgres rejected, a btree index that crashed startup on rows over about 2.7 KB, and fingerprints that changed every scan because of temp-dir paths
- Kept every change verified with 130 backend tests passing on SQLite and PostgreSQL 18 in a 4-job GitHub Actions pipeline, and shipped a non-root Docker image plus a Compose stack (Postgres, API, nginx with WebSocket proxying) at zero ongoing cost

NeuroClass - Brain Tumor MRI Classifier (IEEE 2024) | *Python, ONNX Runtime, FastAPI, Docker, Gradio, Hugging Face, GitHub Actions*

- Served a calibrated MRI classifier at 33 ms CPU inference, 3.5x faster than Keras, by exporting to ONNX Runtime behind a FastAPI endpoint returning class, confidence, and Grad-CAM heatmaps, containerized with Docker and deployed on Hugging Face Spaces
- Rescued inference accuracy from 53 percent to 93.9 percent after a silent collapse toward one class, by isolating the fault to non-converged BatchNorm running statistics and re-estimating them with forward passes that leave the learned weights untouched
- Co-authored the underlying IEEE ICC-ROBINS 2024 publication, and kept the system reproducible and safe with a 10-stage one-command pipeline, 27 pytest tests gating CI, a versioned model card, and a 10 MB upload cap blocking memory-exhaustion attacks

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Java, C/C++, SQL
Backend and APIs: Node.js, Express.js, Fastify, FastAPI, REST APIs, WebSockets, Server-Sent Events, Zod, Pydantic
Frontend: React, Next.js, React Native, Tailwind CSS, React Query
Databases and Storage: PostgreSQL, MySQL, MongoDB, Redis, DynamoDB, Firebase, pgvector, SQLAlchemy, Drizzle ORM
AI and ML: PyTorch, TensorFlow/Keras, Scikit-learn, Transformers, ONNX Runtime, RAG, LangGraph, QLoRA, Hugging Face
Cloud, DevOps and Testing: AWS (EC2, S3, Lambda, RDS), Docker, CI/CD (GitHub Actions), Git, Vercel, Vitest, pytest